

Vishal Kumar Allam

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PROFESSIONAL SUMMARY

- Forward Deployed Engineer and software professional with **5+ years of experience** shipping working systems across insurance, healthcare, and enterprise environments, with recent hands-on experience building **LLM-powered, RAG, and agentic applications** using Python, TypeScript/Node.js, cloud services, APIs, and data platforms.
- Embedded directly with users, consultants, product teams, and technical stakeholders to understand real workflows, translate **ambiguous business problems** into concrete technical solutions, define success metrics, and take ideas from discovery through working prototypes and rapid iteration.
- Designed and demonstrated **agents, internal copilots, document-processing solutions, research and analysis workflows, operational automations, and workflow accelerators** using modern LLMs, tool use, RAG, and agent frameworks including LangGraph, Semantic Kernel, AutoGen, CrewAI, MCP, and A2A concepts.
- Strong backend and full-stack engineering foundation with **Python, FastAPI, Flask, TypeScript, Node.js, React, REST APIs, microservices, SQL, Palantir Foundry/AIP, PySpark, and cloud-native architecture** across AWS and Azure.
- Instrumented AI prototypes with **tracing, evaluations, observability, cost monitoring, latency analysis, model quality metrics, and user feedback loops**, enabling data-driven decisions on whether to iterate, productionize, or stop development.
- Experienced packaging successful prototypes into reusable **components, agent patterns, prompt libraries, technical documentation, architectural notes, evaluation artifacts, known limitations, and handoff materials** for production engineering teams and broader platform reuse.
- Combined strong **business acumen, product sense, client presence, responsible AI, security, compliance, data handling, change management, and commercial awareness** to frame trade-offs, adoption paths, value creation, cost, and implementation feasibility with senior stakeholders.

EDUCATION

University of Missouri Kansas City

"Master of Science" in Computer Science

Kansas City, MO

Jan. 2023 – May 2024

Jawaharlal Nehru Technological University

"Bachelor of Technology" in Computer Science

Hyderabad, India

Aug. 2015 - May 2019

TECHNICAL SKILLS

- **Programming & Application Engineering:** Python, TypeScript, JavaScript, Node.js, FastAPI, Flask, React, C#, Java, SQL, REST APIs, Microservices, Full-Stack Development, Backend Engineering.
- **LLM & Agentic AI:** Generative AI, LLMs, RAG, AI Agents, Agentic Applications, Tool Use, Function Calling, LangGraph, Semantic Kernel, AutoGen, CrewAI, MCP, A2A, Human-in-the-Loop, AI-Assisted Developer Tools, Agentic Coding Workflows.
- **Models & AI Platforms:** OpenAI, Anthropic Claude, Google Gemini, Mistral, Palantir Foundry, Palantir AIP, Prompt Engineering, Embeddings, Vector Databases, Semantic Search, Retrieval Pipelines, Context Engineering.
- **Evaluation & Observability:** LLM Evaluation, Agent Evaluation, Tracing, Model Quality, User Feedback, Cost Monitoring, Latency Monitoring, LLMOps, MLOps, Logging, Observability, Production Diagnostics, Prompt Versioning.
- **Cloud & Data Engineering:** AWS, Azure, GCP Concepts, AWS Lambda, S3, IAM, CloudWatch, Terraform, Docker, PySpark, Spark, ETL/ELT, Data Pipelines, Data Modeling, CDC, PostgreSQL, MySQL, SQL Server.
- **Security & Enterprise Integration:** OAuth2/OIDC, JWT, SAML, RBAC, Identity, Authentication, Authorization, Data Privacy, Responsible AI, Compliance, Secure APIs, Enterprise Integrations, Deployment Patterns.
- **FDE & Product Delivery:** Problem Discovery, User Research, Business Case Framing, Rapid Prototyping, MVPs, Internal Tools, Workflow Automation, Technical Leadership, Client Workshops, Senior Stakeholder Engagement, Architecture Trade-offs, Production Handoff, Documentation, Reusable Accelerators.

EXPERIENCE

Software Engineer AI *PCMI LLC (Full Time)*

Oct 2024 – Present
Dallas, TX, USA

- Embedded with insurance operations, product teams, architects, and end users to identify high-leverage AI opportunities, understand how decisions were actually made, and translate ambiguous workflows into **Palantir, LLM, RAG, and agentic solutions** with clear success metrics and business value.
- Designed and delivered working **prototypes, internal copilots, document-processing agents, research and analysis tools, workflow accelerators, and operational automations** using Palantir Foundry/AIP, Python, FastAPI, LangGraph, OpenAI, Anthropic Claude, Gemini, and enterprise APIs.
- Built agentic applications using **RAG, embeddings, vector databases, tool calling, MCP concepts, multi-step workflows, human-in-the-loop controls, Palantir Ontology, Python services, and governed enterprise data** to execute real insurance and operational work.
- Instrumented prototypes from day one with **traces, evaluations, model-quality metrics, latency, token and infrastructure cost, structured logging, observability, and user feedback**, using results to iterate quickly and determine whether solutions justified further investment.
- Developed reusable **Python/FastAPI components, prompt templates, agent patterns, API connectors, Palantir pipelines, PySpark/SQL transformations, and workflow accelerators**, feeding successful patterns back into shared internal engineering assets.
- Partnered with production engineers to package successful prototypes for clean handoff through **architecture notes, evaluation results, deployment requirements, security and compliance considerations, known limitations, setup guides, and technical documentation**.
- Engaged credibly with senior stakeholders to frame **business cases, adoption paths, revenue and cost implications, technical trade-offs, security, privacy, data realities, scalability, and implementation feasibility**, serving as a trusted technical voice on whether a solution was ready to progress.

Environments: Palantir Foundry, Palantir AIP, Python, FastAPI, LangGraph, Semantic Kernel, OpenAI, Anthropic Claude, Google Gemini, RAG, Vector Databases, MCP, Agentic AI, PySpark, SQL, AWS, Azure, Terraform, Docker, OAuth2/OIDC, JWT, LLM Evaluation, Observability, CI/CD.

Software Engineer AI *Client: CVS*

Dec 2023 – Sept 2024
Remote, USA

- Designed and developed healthcare software and AI-enabled backend services using **Python, FastAPI, Flask, TypeScript, Node.js, REST APIs, AWS, SQL, and microservices**, converting complex operational requirements into working application capabilities.
- Built **LLM-powered internal copilots, RAG prototypes, document-processing workflows, research assistants, and agentic applications** using OpenAI, LangChain/LangGraph patterns, embeddings, vector databases, structured prompts, and healthcare data sources.
- Developed agent workflows that combined **tool use, API integrations, data retrieval, business rules, structured outputs, human review, and asynchronous cloud processing** to automate multi-step healthcare and operational tasks.
- Implemented cloud-native services using **AWS Lambda, S3, IAM, CloudWatch, Docker, event-driven patterns, and secure enterprise APIs**, balancing scalability, identity, deployment complexity, privacy, latency, and operational cost.
- Instrumented AI and backend applications with **structured logging, tracing, latency metrics, usage monitoring, evaluation datasets, prompt iteration, user feedback, and Splunk/CloudWatch observability** to guide rapid product iteration.
- Collaborated with product teams, architects, engineers, and business users through **problem discovery, workflow analysis, technical design, prototype demos, architecture trade-offs, feasibility discussions, and production handoff planning**.
- Created reusable **Python libraries, API clients, prompt patterns, architecture documentation, deployment notes, setup instructions, and operational runbooks**, improving reuse and reducing implementation effort across distributed engineering teams.

Environments: Python, FastAPI, Flask, TypeScript, Node.js, OpenAI, LangChain, LangGraph Concepts, RAG, Embeddings, Vector Databases, AWS Lambda, S3, IAM, CloudWatch, REST APIs, Microservices, Docker, OAuth2/OIDC, JWT, Splunk, LLM Ops, CI/CD.

Software Engineer *Infor*

Sept 2021 – Dec 2022
Hyderabad, India

- Designed and developed enterprise software using **Python, FastAPI, Flask, TypeScript, REST APIs, SQL, and microservices**, supporting internal platforms, operational workflows, and distributed business applications.
- Built reusable backend services and workflow accelerators for **data processing, research-style information retrieval, document handling, business-rule execution, API orchestration, and internal operational automation**.
- Engineered data-intensive solutions using **Python, Pandas, PySpark, SQL, ETL/ELT pipelines, PostgreSQL, MySQL, and Oracle**, transforming incomplete and messy source data into reliable business-facing application outputs.
- Integrated internal and external systems using **REST APIs, JSON, asynchronous processing, event-driven architecture patterns, authentication, reusable API clients, and modular service components**.
- Applied product and engineering judgment to balance **user needs, workflow fit, technical complexity, maintainability, performance, security, and delivery speed**, iterating on solutions based on stakeholder feedback rather than static requirements.
- Supported cloud and platform modernization using **Docker, Jenkins, Git, CI/CD, cloud-native architecture, structured logging, configuration management, and deployment automation** across distributed environments.
- Collaborated with architects, product managers, engineers, and business stakeholders through **problem discovery, technical design, implementation planning, trade-off discussions, documentation, release delivery, and production handoff**.

Environments: Python, FastAPI, Flask, TypeScript, Pandas, PySpark, REST APIs, Microservices, SQL, PostgreSQL, MySQL, Oracle, ETL/ELT, Event-Driven Architecture, Docker, Jenkins, Git, CI/CD, Cloud-Native Applications, Agile/Scrum.

Associate Engineer

June 2019 – August 2021

Virtusa

Hyderabad, India

- Developed enterprise backend services and internal applications using **Python, Flask, REST APIs, SQL, and microservices architecture**, supporting distributed client environments and operational business workflows.
- Built reusable Python components for **workflow processing, data transformation, document and file processing, API communication, business logic, configuration management, logging, and exception handling**.
- Developed RESTful integrations connecting **web applications, backend services, relational databases, external systems, and operational workflows**, using modular APIs and clearly defined service boundaries.
- Engineered data-processing workflows using **Python, SQL, PostgreSQL, MySQL, batch processing, scheduled jobs, and transactional datasets**, improving the reliability and usability of downstream business information.
- Worked directly with senior engineers, architects, analysts, and users to understand **real operational needs, incomplete requirements, workflow constraints, and implementation trade-offs**, then iterated rapidly on practical software solutions.
- Applied secure and maintainable engineering patterns across **authentication, authorization, API contracts, Git, CI/CD, Docker, Linux, structured logging, deployment workflows, and environment configuration**.
- Produced reusable **service components, technical documentation, implementation guides, deployment procedures, operational notes, and handoff materials**, enabling other engineering teams to extend and support delivered solutions.

Environments: Python, Flask, FastAPI Concepts, REST APIs, Microservices, SQL, PostgreSQL, MySQL, JSON, Batch Processing, Event-Driven Architecture, Docker, Git, CI/CD, Linux, Cloud Concepts, API Security, Distributed Systems, Agile/Scrum.

AWARDS & ACHIEVEMENTS

- 2019-11 Oracle Certified Associate, JAVA SE8 Programmer.
- ISTQB Certified Professional.
- Microsoft Technology Associate (MTA) – Database Administration Fundamentals, HTML- 5 Application Development Fundamentals.
- Awarded “Software Process Achievement” Award for my contribution towards the project in Virtusa.
- Received Customer Appreciation mails for quality releases.